Graphical Abstract

HAMFace: Hardness Adaptive Margin Loss for Face Recognition with Various Intra-class Variations

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Highlights

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- Hard Positive Examples.
- Hardness Adaptive Margin Loss.
- SOTA Results 9 Benchmarks.

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HAMFace: Hardness Adaptive Margin Loss for Face Recognition with Various Intra-class Variations

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ABSTRACT

The boost of convolutional neural networks (CNNs) has promoted the development of face recognition. Recently, the emergence of margin-based loss functions further significantly improved the performance of face recognition. These methods, however, sharply degrade in performance when dealing with large intra-class variations, including age, pose, illumination, resolution, and occlusion. Unlike most methods that target concrete variations, our proposed HAMFace address the problems uniformly from the hard positive example perspective. To compress the intra-class variance, we argue that hard positive examples prefer to larger margins which can push them closer to their corresponding class centers. First, an hardness adaptive margin function is designed to adjust the margin according to the hardness of hard positive examples. Then, to improve performance for unconstrained face recognition with various intra-class variations, we propose a novel loss function, named Hardness Adaptive Margin (HAM) Softmax Loss, which allocates larger margins to hard positive examples during training according to their hardness. The proposed HAMFace model is evaluated on nine challenging face recognition benchmarks and exhibits its superiority compared with other state-of-the-arts.

1. Introduction

Progress in deep CNNs (He et al., 2016) has significantly improved the face recognition performance (Schroff et al., 2015; W. Liu et al., 2017). The margin based loss methods (W. Liu et al., 2017; H. Wang et al., 2018; Deng, Guo, Xue, & Zafeiriou, 2019) are the most popular approaches for application, which exhibit high accuracy in general face recognition. However, it offen suffers degradation in cases of challenges due to large varying in age, pose, resolution, and occlusion (Tripathi & Jalal, 2021; Cao et al., 2016; Z. Zhang et al., 2015; Wen et al., 2016; Masi et al., 2018).

To alleviate this issue, previous studies have primarily focused on dealing with a concrete variation, such as cross-pose variation(Masi et al., 2018; Marriott et al., 2021) and cross-age variation (Y. Wang et al., 2018; Zhao et al., 2019; Wu et al., 2021). To deal with the imbalanced distribution of concrete variation, researchers resort to synthesizing face images to augment the training dataset (Yin et al., 2017; Marriott et al., 2021). Despite the success in face recognition with concrete variations, such as the cross-pose (Masi et al., 2018; Marriott et al., 2021) and cross-age recognition (Y. Wang et al., 2018; H. Wang et al., 2019) tasks, the performance degrades when a single model is applied to multiple variations. Alternatively, models trained individually on concrete variation datasets are integrated to accommodate hybird variations but at the cost of additional runtime complexity (Masi et al., 2018). Recently, Shi et al. (Shi et al., 2020) proposed a novel framework to learn universal features of face recognition by associating them with different variations, which achieved good performance on both general face recognition datasets and extreme benchmarks. That said, such a method has a shortcoming in that it relies heavily on data augmentation for concrete variations, but some potential variations in the dataset may not be clearly defined.

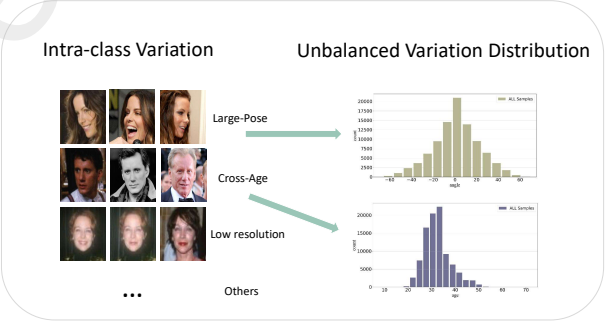

Figure 1: Intra-class variation and imbalanced data distribution are the common challenges of unconstrained face recognition with large pose, cross-age, low resolution and so on.

Rethinking the learning of invariant face representation, we focus on improving model robustness from the hard positive example perspective to tackle the challenges of various intra-class variations in unconstrained environments. As shown in Figure 1, intra-class variation and the imbalanced data distribution are common challenges in unconstrained face recognition. It can be observed that different variations of face images may be mixed together in the unconstrained scenarios. These motivate us to address the problem of various concrete variations of face recognition from a unified perspective.

Indeed, the distributions of various kinds of variations are unevenly distributed in the face datasets, even within individuals (C. Liu et al., 2022). In addition, individual face variations often result in significant intra-class variance that exerts a negative impact on face recognition. Both of these

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lead to model performance degradation in unconstrained scenarios. The margin based methods (W. Liu et al., 2017; H. Wang et al., 2018; Deng, Guo, Xue, & Zafeiriou, 2019) introduce a uniform margin to separate different classes and tend to achieve intra-class compactness indirectly, while ignoring the imbalanced distribution of intra-class variations. This, to some extent, hinders performance in scenarios with large intra-class variations.

Recently, researchers have turned their attention to adaptive margins for improving face recognition performance, including class-level adaptive margin (H. Liu et al., 2019) and sample-level adaptive margin (Meng et al., 2021; Kim et al., 2022). The class-level adaptive margin (H. Liu et al., 2019) shares the same importance within the same indentity. In contrast, the finer-scale sample-level margin can differ samples from the same person in their importance. MegFace (Meng et al., 2021) adopted larger margins to high-quality samples and pulled the easy samples towards their corresponding centers while pushing hard samples away, achieving good performance on mixed and low-quality face datasets. They measured the face quality based on the magnitude of feature embeddings. AdaFace (Kim et al., 2022) further demonstrated that feature norms can be used as the proxy image quality. They assigned different importance to different samples according to their image quality whilst hard samples with low image quality tend to be ignored. However, although the feature norms are related to image quality to some extent, they do not accurately reflect the image quality. Likewise, the image quality does not always accurately reflect the hardness of a sample.

In this paper, we propose a novel Hardness Adaptive Margin (HAM) loss which directly allocates larger margins to hard positive examples according to their hardness. The distribution of hard positive examples in dataset is quite long-tailed, which is positively correlated with the imbalanced intra-class variations. From the hard positive examples perspective, we utilize larger margins to reduce intra-class variance according to their hardness, aiming to promote the discriminative power in unconstrained face recognition with large intra-class variations. We argue that allocating larger margins adaptive to the hardness of hard positive examples is beneficial to handling imbalanced distribution of intra-class variations. In our prior research (Xiao et al., 2022), a hard example mining method that explicitly identifies hard positive examples during the training has been proposed. This makes it possible to focus on hard positive examples which are crucial to the learning objective of intra-class compactness. Therefore, we firstly design an adaptive margin function to allocate hard positive examples with larger margins according to their hardness. Then, we propose a novel hardness adaptive margin loss to uniformly cope with the challenges for face recognition with large intra-class variations in unconstrained scenarios. A large margin that fits the hardness of the example acts like a large wedge to push the hard example toward the corresponding class center.

The major contributions of this work are as follows:

- We design an adaptive margin function to adjust margin directly according to the hardness of the hard positive examples.
- To refine discriminative features, we design a novel Hardness Adaptive Margin loss, which allocates larger margins to the hard positive examples and push them close to their corresponding class centers, thus improving the intra-class feature distribution.
- Our proposed HAMFace is evaluated on nine face recognition benchmarks, experimental results of which show that it achieves top performance in unconstrained face recognition even with various intra-class variations.

2. Related Work

2.1. Margin-based Loss Fucntion

Margin-based losses (W. Liu et al., 2017; H. Wang et al., 2018; Deng, Guo, Xue, & Zafeiriou, 2019) have brought improved discriminative performance for unconstrained face recognition. To improve discriminative power of features, L-Softmax (W. Liu et al., 2016) was proposed to explicitly encourage larger inter-class variance by imposing an margin between classes. SphereFace (W. Liu et al., 2017) further improved L-Softmax by introducing normalization on weights and obtained SOTA results on LFW (G. B. Huang et al., 2008),YTF (Wolf et al., 2011) and MegaFace (Kemelmacher-Shlizerman et al., 2016). This is a milestone in deep face recognition. From then on, variants of SphereFace, such as CosFace (H. Wang et al., 2018) and ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019), employed margin penalty in different manners to enhance face recognition performance while also stabilizing the training process. Wang et al. (H. Wang et al., 2018) normalizd both weight vectors and features, and then maximized the decision margin to learn highly discriminative features by introducing a cosine margin. ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019) introduced an additive angular margin to the target angle, which provided a clear geometric interpretation and achieved SOTA results on over 10 challenging benchmarks.

The margin based loss are formulated as:

$$\mathcal{L} = -\frac{1}{N}\sum_{i}\log\frac{exp\left(s\cdot\psi\left(\theta_{y_i},m\right)\right)}{exp\left(s\cdot\psi\left(\theta_{y_i},m\right)\right)+\sum_{j\neq y_i}^{n}exp\left(s\cdot\cos\left(\theta_j\right)\right)}, \tag{1}$$

where s denotes the hyper-parameter for scaling. m refers to the margin. y_i refers to the index of the ground truth label. θ_j refers to the angle between sample feature and the weight vector of jth class. $\psi\left(\theta_{y_i,m}\right)$ refers to the margin function, it can be formulated as follows:

$$\psi\left(\theta_{y_i,m}\right)_{SphereFace} = \cos\left(m\cdot\theta_{y_i}\right), \tag{2}$$

$$\psi\left(\theta_{y_i,m}\right)_{CosFace} = \cos\left(\theta_{y_i}\right) - m, \tag{3}$$

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$$\psi\left(\theta_{y_i,m}\right)_{ArcFace} = \cos\left(\theta_{y_i} + m\right). \quad (4)$$

Although the above-mentioned methods (W. Liu et al., 2017; H. Wang et al., 2018; Deng, Guo, Xue, & Zafeiriou, 2019) have achieved good performance, the uniform margin ignoring the long-tailed distribution of dateset, such as classes and various variations, limits the performance in practical applications to some extent.

2.2. Face Recognition with Concrete Variations

LF-CNN (Wen et al., 2016) was proposed to handle cross-age face recognition. The age-invariant face features were first extracted by coupling the CNNs parameters and the latent identity analysis (LIA) module parameters. Then, OE-CNNs (Y. Wang et al., 2018) was adopted for learning age-invariant components from face features. Face features were categorized into two orthogonal coordinate systems and supervised by different signals to learn identity-related features and age-related features, respectively. Wang et al. (H. Wang et al., 2019) introduced a decorrelated adversarial learning (DAL) method for removing age-related components from face features, which is the first study of its kind. Zhao et al. (Zhao et al., 2019) introduced an Age-Invariant Model (AIM), which utilized a unified CNN architecture to jointly perform cross-age recognition in a mutually beneficial manner. Effective training strategies were used for end-to-end learning to separate robust age-invariant. Wu et al. (Wu et al., 2021) proposed the PMADN to cope with the limitations of existing cross-age face recognition methods. To obtain age-robust features, they adopted parallel multipath full connected layers to map features into different age groups, and then performed nonlinear recombination of them.

PAM (Masi et al., 2018) was proposed to improve robustness of model to pose variations in face verification. They utilized 3D rendering to synthesize multiple poses from input data and used to pose-specific models training. 3D GAN (Marriott et al., 2021) was proposed to learn a nonlinear texture model, augmenting dataset with synthetic images to improve the performance of large-pose face recognition.

To mitigate model performance degradation in case of wide variations, Shi et al. (Shi et al., 2020) introduced a universal representation learning method for face recognition. They splited the feature embedding into multiple sub-embeddings related to different variations, and further decorrelated the sub-embeddings by regularizing variation classification loss and variation adversarial loss.

These approaches for face recognition under concrete variations always rely on face synthesis to augment the training dataset. However, data augmentation inevitably introduces artifacts, which are detrimental to the training process. Moreover, some undefined potential variations may exist in the datasets. Unlike variation-specific approaches, our method copes with the challenge of large intra-class variation face recognition from a higher dimensional perspective.

2.3. Adaptive Margin Methods

To reduce the bias of long-tailed classes, AdaM-Softmax (H. Liu et al., 2019) was proposed to adaptively assign different margins to different classes base on the class volume. It introduced hard prototype mining to adaptively select some hard prototypes and concentrate on hard classes. MV-Softmax (X. Wang et al., 2020) was further proposed to utilize the misclassified vectors to guide the training process. It adopted hard example mining to identify hard samples and differentiate them from classes which have no ground truth. Based on this, CurricularFace (Y. Huang et al., 2020) further introduced an adaptive curriculum learning loss to identify easy examples first before focusing on hard examples during training.

In addition, many studies have adopted adaptive margins based on image quality (Meng et al., 2021; Kim et al., 2022). MagFace (Meng et al., 2021) adopted feature norm as the indicator of image quality and used large margins for easier identification of features. As a result, easy examples were pulled closer to their corresponding centers whereas hard examples with low feature norms were pushed away. Kim et al. (Kim et al., 2022) further demonstrated that feature norm is strongly correlated with image quality. They utilized feature norm as the proxy for image quality and proposed AdaFace to adjust the importance of examples of different hardness according to feature norms. They focused on hard but recognizable examples rather than unrecognizable ones. However, feature norm cannot accurately reflect the hardness of an example. Different from (Meng et al., 2021; Kim et al., 2022), our proposed HAMFace directly allocates larger margins to hard positive examples according to their hardness. The larger margin assigned according to hardness push hard example closer to its corresponding center.

3. Proposed METHOD

In order to improve intra-class distribution, we employ larger margin to push the hard sample closer to its corresponding centers. Figure 2 show the architectural overview of the proposed HAMFace. This section starts with a description of our motivation for conducting the study. It continues to introduce an adaptive margin method which assigns margins based on the hardness of hard positive examples. In the end, a novel Hardness Adaptive Margin Loss to emphasize hard positive examples during training is presented, aiming to uniformly cope with the challenges for unconstrained face recognition with large intra-class variations.

3.1. Motivation

Much scholarly attention has thus far been paid to learning invariant face representations with various concrete variations in real-world scenarios. However, we shift our focus to intra-class variation, which is a common barrier to improving face recognition in a variety of concrete variations. Intuitively, improving the intra-class compactness is beneficial to improving the unconstrained face recognition performance.

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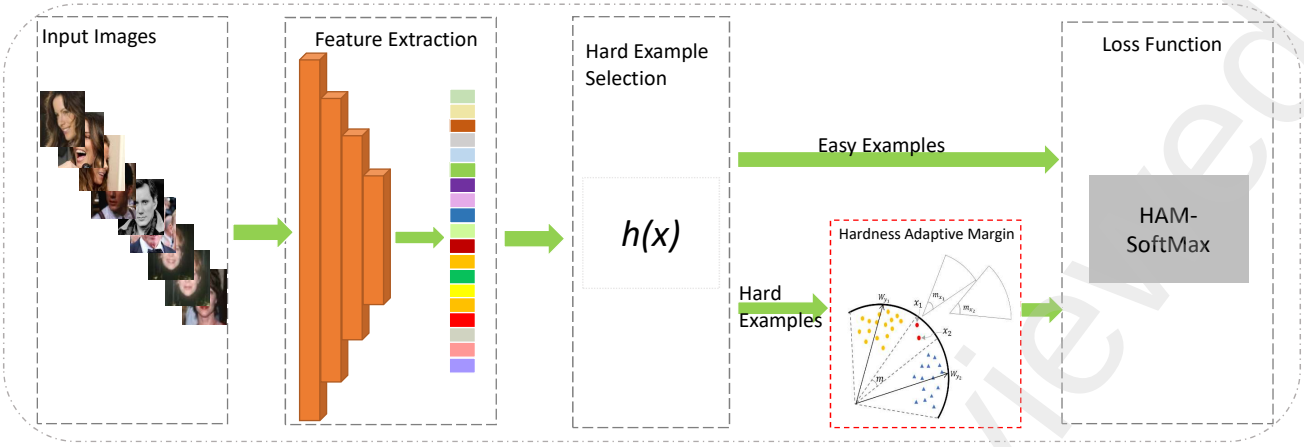

Figure 2: Overview of the proposed HAMFace architecture.

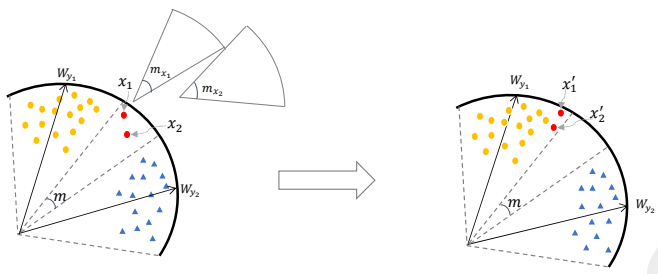

Figure 3: Illustration of hard sample feature x adaptively be assigned with a larger margin. W_{y_i} servers as a center for the class of the ground truth y_i. m refers to the fixed margin. m_{x_i} refers to the adaptive larger margin for hard sample x_i. x_i' denotes the position of x_i at the end of training procedure.

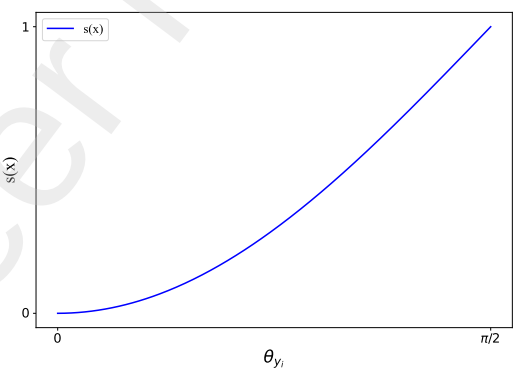

Figure 4: The curve of $s(x)$, θ_{y_i} range from 0 to $\pi/2$.

Given the common occurrence of hard positive samples and their imbalanced distribution, it is hard to achieve intra-class compactness with uniform margins of softmax-based methods, especially for individuals with large intra-class variations in unconstrained scenarios. Some recent studies (H. Liu et al., 2019; Kim et al., 2022,?) have achieved good performance by integrating adaptive margins to margin-based loss for face recognition. These motivate us to utilize adaptive margins and focus on hard positive examples during training. Specifically, to improve the intra-class compactness, we prefer to adopt larger margins for hard positive examples directly according to their hardness. Larger margin that corresponds to harde positive example function acts like a wedge, pushing the hard positive example closer to its corresponding center. Figure 3shows that sample feature x_1 and x_2 are assigned larger margin m_{x_1} and m_{x_2} according to their hardness, respectively. m_{x_1} and m_{x_2} act as wedges to squeeze x_1 and x_2 close to W_{y_1} during training.

3.2. Hardness Adaptive Margin Function

Previous studies have revealed that hard positive examples are important for improving face representations (H. Liu et al., 2019; X. Wang et al., 2020; Kim et al., 2022). Assigning hard classes with larger margins has been proved to be effective to reducing intra-class variations (H. Liu et al., 2019).

Large intra-class variance is primarily attributed to wide existence of hard positive examples. Thus, we should first consider how to identify hard positive examples. In our prior work (Xiao et al., 2022), a hard example selection function has been introduced which can effectively identify hard positive examples based on margin based loss. In case of Arcface, the function was formulated as:

$$h(x) = \begin{cases} 0 & \text{if } \theta_{y_i} + m \leq \min(inter) \\ 1 & \text{if } \theta_{y_i} + m > \min(inter) \end{cases}, \tag{5}$$

where θ_{y_i} refers to the angle between sample x and the ground truth weight W_{y_i}, $min(inter)$ denotes the minimum angle between sample x and its non-ground truth weight vectors, parameter m refers to the fixed angular margin.

We prefer to adaptively allocate larger margins to hard positive examples, aiming to push them closer to their class centers. θ_{y_i} acts as the indicator for the hardness of example. The hardness of example is positively associated with the value of θ_{y_i}. We also design a function $s(x)$ to scale the

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margins of hard examples according to their hardness. It can be formulated as:

$$s(x) = 1 - \cos\theta_{y_i}, \tag{6}$$

where function $s(x)$ is convex increasing from 0 to $\pi/2$, as shown in Figure 4. It can be observed that $s(x)$ increases slowly when the value of θ_{y_i} is close to 0 but more rapidly when the value of θ_{y_i} approaches $\pi/2$. This implies that harder example would be allocated with larger margins.

We further define the hardness adaptive margin function such that the margin for hard example is consistently greater than the fixed margin and increases with the hardness of example. The hardness adaptive margin function is formulated as:

$$m(x) = m + t \cdot h(x) \cdot s(x), \tag{7}$$

where t is a scaling factor that adjusts $m(x)$ to be within an appropriate range, and parameter m refers to a fixed angular margin. $m(x)$ becomes the fixed margin m when t is 0. We set t to a small value to avoid the overall performance degradation and convergence failure caused by excessively large margins $m(x)$. It can also be observed that $m(x)$ becomes the fixed margin while $h(x)$ equals 0. That means easy examples are allocated with the fixed margin m, and the number of them are overwhelmingly during training process.

3.3. Hardness Adaptive Margin Loss

Recent studies (H. Liu et al., 2019; Meng et al., 2021; Kim et al., 2022) on face recognition suggest that applying adaptive margin to margin-based loss is effective to enhance the discriminative performance in unconstrained face recognition. In this section, we propose a Hardness Adaptive Margin (HAM) Softmax loss, that focuses on hard positive examples to uniformly cope with the challenges for unconstrained face recognition with various intra-class variations. To improve intra-class compactness, larger margins are allocated to hard positive examples directly according to their hardness, thereby pushing them towards their corresponding class centers. The proposed HAM loss can be defined as:

$$\mathcal{L}_{HAM} = -\frac{1}{N} \cdot \sum_i \log \frac{exp\left(s \cdot \cos\left(\theta_{y_i} + m(x)\right)\right)}{exp\left(s \cdot \cos\left(\theta_{y_i} + m(x)\right)\right) + \sum_{j \neq y_i} exp\left(s \cdot \cos\left(\theta_j\right)\right)}, \tag{8}$$

where s is the parameter for scaling. The HAM loss takes two different forms, with one used for easy samples and the other for hard samples. When θ_{y_i} is smaller than $min(intra)$, the sample is identified as an easy example, and the HAM loss for easy examples is defined as:

$$\mathcal{L}_{HAM-E} = -\frac{1}{N} \cdot \sum_i \log \frac{exp\left(s \cdot \cos\left(\theta_{y_i} + m\right)\right)}{exp\left(s \cdot \cos\left(\theta_{y_i} + m\right)\right) + \sum_{j \neq y_i} exp\left(s \cdot \cos\left(\theta_j\right)\right)}, \tag{9}$$

where m is a fixed angular margin. We can see that HAM loss becomes ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019) for easy examples, which dominates the training process. By contrast, when θ_{y_i} is greater than $min(intra)$, the sample is identified as a hard positive example. We can substitute Eq. 5 and Eq. 6 into Eq. 7, the adaptive margin function for hard positive examples can be formulated as:

$$m_h(x) = m + t \cdot \left(1 - \cos\theta_{y_i}\right), \tag{10}$$

Thus, the HAM loss for hard positive examples can be defined as:

$$\mathcal{L}_{HAM-H} = -\frac{1}{N} \cdot \sum_i \log \frac{exp\left(s \cdot \cos\left(\theta_{y_i} + m_h(x)\right)\right)}{exp\left(s \cdot \cos\left(\theta_{y_i} + m_h(x)\right)\right) + \sum_{j \neq y_i} exp\left(s \cdot \cos\left(\theta_j\right)\right)}. \tag{11}$$

Indeed, easy examples far outnumber hard positive examples and dominate the training process. During model training, the easy examples are assigned the fixed margin m whereas the hard positive examples are assigned adaptive larger margins according to their hardness. Therefore, larger margin push the hard positive example towards its corresponding class center and minimize the intra-class variance consequently.

Figure 5 visualizes θ_{y_i} and margin distribution of hard positive examples. From Figure 5, it can be observed that θ_{y_i} is long-tailed, and the margin changes corresponding to θ_{y_i}. The head of θ_{y_i} ranges from 35° to 55° and it peaks at around 40°. It can also be observed from Figure 5(b) that the peak value of margin is around 0.55, and there are very few margins greater than 0.70.

4. EXPERIMENTS

4.1. Datasets

CASIA-Webface (Yi et al., 2014) and MS1MV3 (Deng, Guo, Zhang, et al., 2019) are used as our training datasets. Models are evaluated on nine popular benchmarks, as shown in Table 1. To verify the effectiveness of our HAM loss in scenarios with large intra-class variations, we conduct experiments on several cross-age and cross-pose benchmarks, even including hybrid variations benchmarks. AgeDB-30

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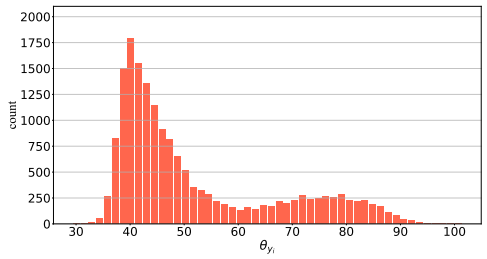
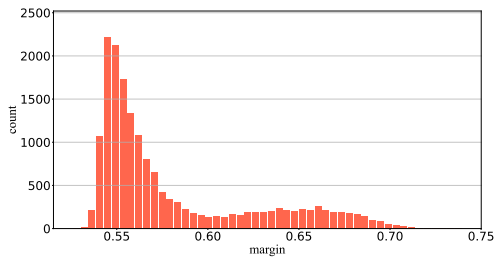


(a) The distribution of θ_{y_i} for hard positive examples.



(b) The distribution of margin for hard positive examples.

Figure 5: Visualizion the θ_{y_i} and margin distribution for hard positive examples. Hard positive examples of 2000 individuals randomly sampled from CASIA-Webface.

Table 1
DATASETS FOR TRAINING AND TESTING.

#Date Set		#Identity	#Image
Training	CASIA-Webface	10k	500k
	MS1MV3	91k	5.1M
Test	AgeDB-30	568	16,488
	CALFW	5,749	12,174
	CACD-VS	2,000	163,446
	CFP-FP	500	7,000
	CPLFW	5,749	11,652
	IJB-B	1,845	76,824
	IJB-C	3,531	148,876
	LFW	5,749	13,233
	MegaFace	530(p)	1M(g)

(Moschoglou et al., 2017), CALFW (Zheng et al., 2017) and CACD-VS (Chen et al., 2015) are used for cross-age face recognition, while CFP-FP (Sengupta et al., 2016) and CPLFW (Zheng & Deng, 2018) serve for cross-pose face recognition. IJB-B (Whitelam et al., 2017) and IJB-C (Maze et al., 2018) are two popular hybrid variant benchmarks. We also verify the effectiveness of HAM loss on general face recognition benchmarks, including LFW (G. B. Huang et al., 2008) and MegaFace (Kemelmacher-Shlizerman et al., 2016).

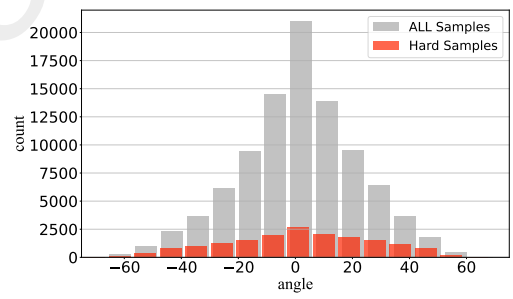
4.2. Experimental Settings

Following (Deng, Guo, Xue, & Zafeiriou, 2019; K. Zhang et al., 2016; Deng et al., 2020), images are cropped to 112×112, aligned with five landmarks. ResNet (He et al., 2016) is employed as the network architecture of different depths, including ResNet34, ResNet50 and ResNet100. The models are trained with SGD algorithmh on four TITAN RTX GPUs parallelly, with the batch size of 512. Learning rate is initialized to 0.1, which is then divided by 10 at 26K and 40K iterations on CasIA-WebFace, and by 10 at 100K, 150K and 180K iterations on MS1MV3.

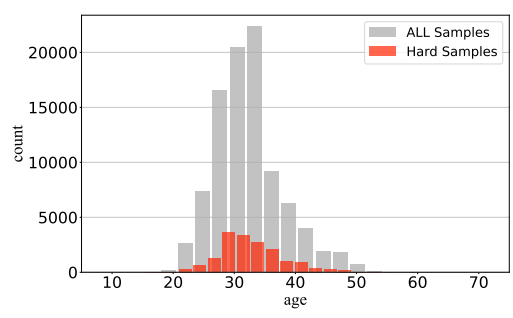
4.3. Visualization of Hard Example Distribution

The distributions of faces by pose and age are visualized in Figure 6. As seen from Figure 6, their distributions are highly uneven. The facial poses are mainly distributed bewteen $-30°$ to $+30°$, while the ages vary between 22 and 42. Obviously, the distribution of poses differs greatly from that of age groups for hard positive examples. The pose distribution of hard positive examples is flatter than the age distribution. To some extent, this indicates that large pose variations exert more impact on the distribution of hard positive examples. Indeed, the number of samples with pose variations far exceeds that of samples with age variations in training dataset.

Figure 7 shows the hard positive examples chosen from the CASIA-Webface dataset. Faces of eight individuals, categorized as hard positive examples, feature various intra-class variations, such as age, pose, occlusion, and resolution.



(a) Pose distribution



(b) Age distribution

Figure 6: Pose and age distribution of 2000 individuals randomly sampled from CASIA-Webface.

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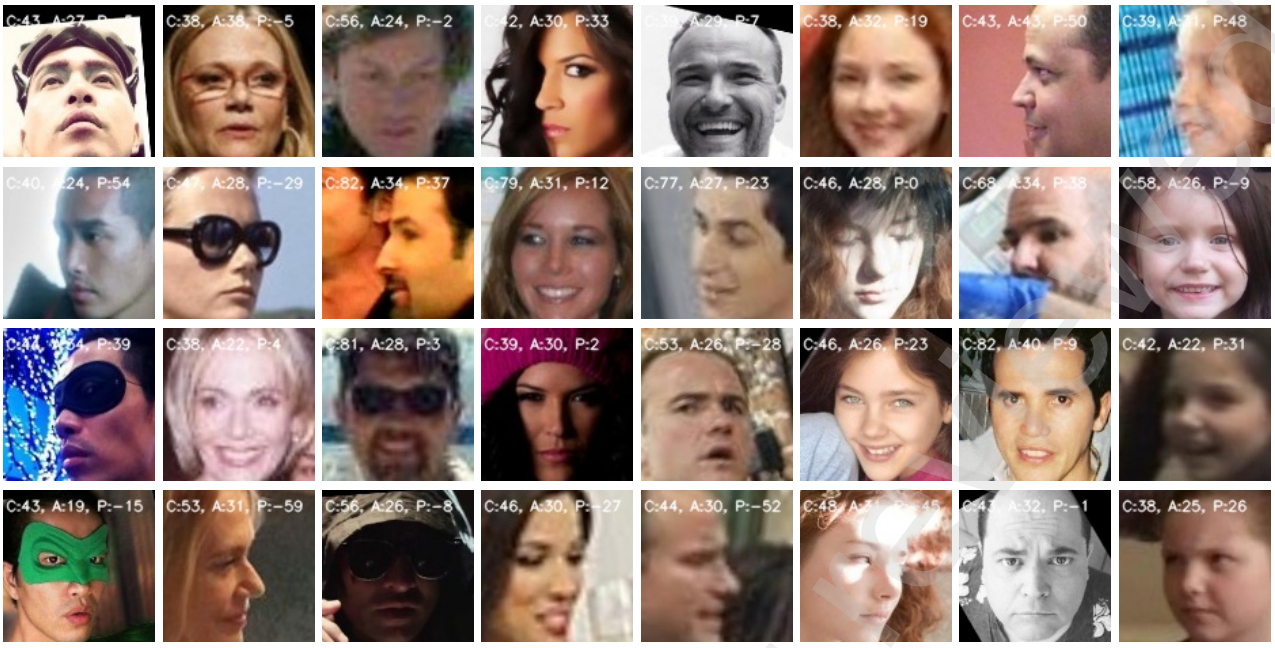

Figure 7: Visualization of hard positive examples that are randomly selected from the CASIA-Webface dataset. Each column of face images belongs to the same identity. The "C" denotes the value of θ_{y_i}. The "A" refers to the age. The "P" refers to the angle of face pose.

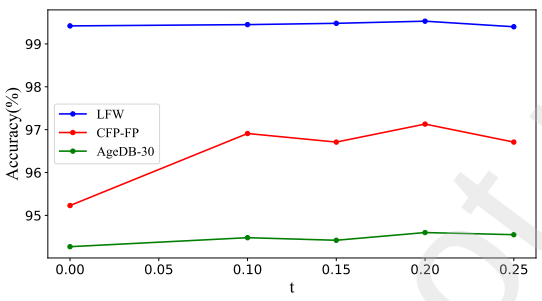

Figure 8: The influence of t on three benchmarks.

Table 2
STUDY ON HYPERPARAMETER t FOR HAMFACE.

t	LFW	CFP-FP	AgeDB-30
0	99.42	95.23	94.27
0.10	99.45	96.91	94.48
0.15	99.48	96.71	94.42
0.20	**99.53**	**97.13**	**94.60**
0.25	99.40	96.71	94.55

Table 3
STUDY ON HYPERPARAMETER m.

m	LFW	CFP-FP	AgeDB-30
0.4	99.43	96.61	94.58
0.45	99.45	96.92	94.30
0.5	**99.53**	**97.13**	**94.60**
0.55	99.50	97.01	94.20

4.4. Ablation Study

4.4.1. Effect of Hyperparameter t

In this section, we investigate of Hyperparameter t in the proposed HAM loss. We train our ResNet34 model on CASIA-WebFace (Yi et al., 2014) with different parameters of t. We conduct the evaluation with different values of t which range from 0 to 0.25. The proposed HAMFace becames ArcFace when t is 0. As shown in Table 2, the proposed model achieves better performance when t is greater than zero, and reaches its saturation point when t equals 0.20. From Figure 8, we can observe that the improvements on the three benchmarks are obviously, especially on CFP-FP.

4.4.2. Effect of Hyperparameter m

The margin parameter m is very import to HAM loss. We examine the effect of Hyperparameter m in the proposed HAMFace. The Resnet34 model is trained on CASIA-WebFace (Yi et al., 2014). We evaluate its performance on LFW (G. B. Huang et al., 2008), CFP-FP (Sengupta et al., 2016) and AgeDB-30 (Moschoglou et al., 2017) by varying m from 0.4 to 0.55. Table 3 shows the evaluation results on the three benchmarks. It can be observed that as the hyperparameter m increases in value, the accuracy of HAMFace becomes better, and achieves the best results on these three benchmarks when m equals 0.5. In the subsquent experiments, m is set to 0.5.

4.4.3. Study of Semi-hard Example

To study the effect of semi-hard example, we add constraints to θ_{y_i} in Eq. 5, with the number varying from 40° to 80°. As previously mentioned, We train the proposed HAM-

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Table 4
STUDY ON SEMI-HARD EXAMPELS. "Avg" REFERS TO THE AVERAGE PERFORMANCE(%) ON LFW, CFP-FP, AND AGEDB-30. "-" DENOTES NO LIMITATION FOR θ_{y_i}.

Range	LFW	CFP-FP	AgeDB-30	Avg
$\theta_{y_i} < 40°$	99.48	96.63	93.73	96.61
$\theta_{y_i} < 50°$	99.50	97.07	94.47	97.01
$\theta_{y_i} < 60°$	**99.58**	**97.27**	94.47	**97.11**
$\theta_{y_i} < 70°$	99.55	97.06	94.20	96.94
$\theta_{y_i} < 80°$	99.52	97.14	94.00	96.88
–	99.53	97.13	**94.60**	97.09

Face on CASIA-WebFace with ResNet34 and evaluate it on LFW, CFP-FP and AgeDB-30. Table 4 shows the results with different constraints on θ_{y_i}. We can see that the proposed HAMFace gets saturation on LFW, CFP-FP when θ_{y_i} is smaller than 60°. Compared with the baseline which do not consider semi-hard examples, the performance of HAMFace improves by 0.05% on LFW and 0.14% on CFP-FP respectively, while decreases by 0.13% on AgeDB-30. On average, the performance improves by 0.02%.

4.5. Evaluation Results

4.5.1. Evaluation on Cross-Pose Benchmarks

We first evaluate our method on cross-pose benchmarks, such as CFP (Sengupta et al., 2016) and CPLFW (Zheng & Deng, 2018). CFP (Sengupta et al., 2016) is a challenging dataset for cross-pose face verification, containing faces of 500 celebrities. Each identity consists of 10 frontal and 4 profile faces. The standard protocol (Sengupta et al., 2016) is used for evaluation. CPLFW dataset (Zheng & Deng, 2018) contains 11k face images of 5.7k identities with various pose variations.

We train our HAMFace model on MS1MV3 with ResNet100 backbone. Table 5 shows that our proposed HAMFace achieves optimal performance on CFP-FP and outperforms all other competitors on CPLFW except for 3D GAN(FFHQ) (Marriott et al., 2021). However, 3D GAN(FFHQ) is an advanced approach for cross-pose face recognition. Compared to ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019), the proposed HAMFace gains 0.79% performance improvement on CFP-FP and 1.25% on CPLFW. Compared with MvCoM-URFace (C. Liu et al., 2022), our model performs better by 0.59% on CFP-FP and by 0.47% on CPLFW. The proposed HAMFace is also competitive to the approaches for cross-pose face recognition. Compared with URL (Shi et al., 2020), our HAMFace gains 0.42 % on CFP-FP. Compared to 3D GAN(FFHQ) (Marriott et al., 2021), the proposed method improves by 0.55% on CFP-FP whereas drops by 0.20% on CPLFW.

4.5.2. Evaluation on Cross-Age Benchmarks

We evaluate our model on AgeDB-30 (Moschoglou et al., 2017), CALFW (Zheng et al., 2017) and CACD-VS (Chen et al., 2015). AgeDB-30 dataset (Moschoglou et al., 2017) is a manually collected database for cross-age face verification, consisting of 16k images of 568 identities. CALFW (Zheng et al., 2017) is a variant of LFW (G. B. Huang et al., 2008), which is designed for face verification with large cross-age varation. CACD (Chen et al., 2015) is a challenging cross-age face recognition benchmark, consisting of 163k faces of 2k subjects with various variations in age, pose, and illumination. CACD collects data from the Internet and thus contains many mislabeled and duplicate faces. CACD-VS (Chen et al., 2015) is a carefully annotated version of CACD, including 2k positive pairs and 2k negative pairs.

We evaluate our ResNet100 model on CALFW and AgeDB-30, and further compare it with the SOTA approach. Table 6 shows that our proposed HAMFace maintains a consistent advantage over its competitors on both CALFW and AgeDB-30. To be specific, HAMFace outperforms ArcFace by 0.82% on CALFW and 0.30% on AgeDB-30. Compared with AdaFace (Kim et al., 2022), our model gains an improvement of 0.25% in performance on CALFW and 0.28% on AgeDB-30. HAMFace also has a slight edge over CurricularFace (Y. Huang et al., 2020) with a performance improvement of 0.07% on CALFW and 0.13% on AgeDB-30.

Table 5
VERIFICATION PERFORMANCE(%) ON CFP-FP and CPLFW.

Method	CFP-FP	CPLFW
SphereFace (W. Liu et al., 2017)	-	81.40
CosFace (H. Wang et al., 2018)	98.12	92.28
ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019)	98.27	92.08
MV-Arc-Softmax (X. Wang et al., 2020)	95.70	89.69
CurricularFace (Y. Huang et al., 2020)	98.37	93.13
MagFace (Meng et al., 2021)	98.46	92.87
MvCoM-URFace (C. Liu et al., 2022)	98.47	92.86
IHEM (Xiao et al., 2022)	99.03	93.25
URL (Shi et al., 2020)	98.64	-
3D GAN(FFHQ) (Marriott et al., 2021)	98.51	**93.53**
HAMFace(ours)	**99.06**	93.33

Table 6
ACCURACY (%) ON CALFW AND AGEDB-30.

Method	CALFW	AgeDB-30
SphereFace (W. Liu et al., 2017)	90.30	-
CosFace (H. Wang et al., 2018)	95.76	98.11
ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019)	95.45	98.15
MV-Arc-Softmax (X. Wang et al., 2020)	95.63	98.11
CurricularFace (Y. Huang et al., 2020))	96.20	98.32
MagFace (Meng et al., 2021)	96.15	98.17
AdaFace (Kim et al., 2022)	96.02	98.17
IHEM (Xiao et al., 2022)	96.20	98.37
HAMFace(ours)	**96.27**	**98.45**

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Table 7
ACCURACY (%) ON CACD-VS.

Method	ACC(%)
LF-CNN (Wen et al., 2016)	98.50
OE-CNN (Y. Wang et al., 2018)	99.20
AIM (Zhao et al., 2019)	99.38
DAL (H. Wang et al., 2019)	99.40
HAMFace(ours)	**99.55**

To further compare with cross-age face recognition methods, we evaluate our ResNet50 model on CACD-VS. Table 7 shows that our proposed HAMFace achieves competitive result on CACD-VS dataset (Chen et al., 2015) compared with existing cross-age face recognition approaches.

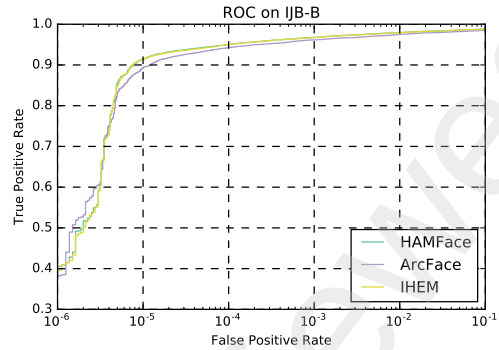
4.5.3. Evaluation on Hybrid Variations Benchmarks

We conduct experiments on hybrid intra-class variations benchmarks, including IJB-B (Whitelam et al., 2017) and IJB-C (Maze et al., 2018). Different variations, such as large occlusions, low image quality, and diverse illumination conditions, get mixed in the dataset, which constitutes the major challenge of these two benchmarks. The IJB-B (Whitelam et al., 2017) dataset contains 21.8K images and 55K frames belonging to 1,845 subjects. As an extended version of IJB-B, the IJB-C (Maze et al., 2018) dataset contains 31k images and 117k frames belonging to 3,500 identities.

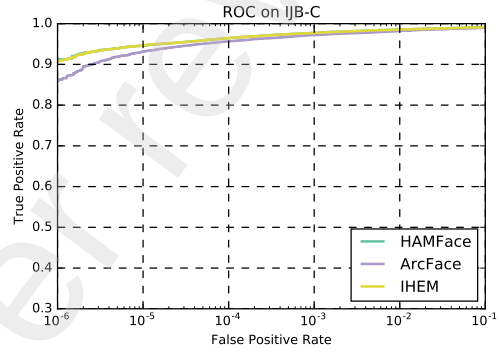
As stated earlier, HAMFace model is trained on MS1MV3. Table 8 clearly demonstrates the superiority of our HAMFace on these two hybrid variations Benchmarks. Compared to ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019), the proposed HAMFace achieves significant improvement in all FAR criteria, gaining an improvement of 2.33%, 2.23%, 0.79% on IJB-B at FAR=1e-6, FAR=1e-5, FAR=1e-4 and 5.0%, 1.47%, 0.81% on IJB-C. Compared with MagFace (Meng et al., 2021), HAMFace performs better by 1.20%, 0.53% on IJB-B at FAR=1e-5, FAR=1e-4 and 1.01%, 0.54%, 0.49% on IJB-C at FAR=1e-6, FAR=1e-5, FAR=1e-4. In comparison with IHEM (Xiao et al., 2022), HAMFace shows improvements in performance of 0.82%, 0.19% on IJB-B at FAR=1e-6, FAR=1e-5 and 0.32% on IJB-C at FAR=1e-6, although it experiences a decline of 0.06% on IJB-C at FAR=1e-5. The ROC curves are shown in Figure 9. We can see from Figure 9 that the improvement of HAMFace is impressive compared to ArcFace.

4.5.4. Evaluation on General Benchmarks

To further verfiy the generalizability of the proposed HAMFace, we evalute it on two general face recognition benchmarks, including LFW (G. B. Huang et al., 2008) and MegaFace Challenge 1 (Kemelmacher-Shlizerman et al., 2016). LFW (G. B. Huang et al., 2008) is widely used benchmark for face recognition, consisting of 13k images of 5.7k identities with various poses, ages, facial expressions and illuminations in an unconstrained environment. The MegaFace dataset includes a gallery set and a probe set, with a total 1M face images in the former and 100K face images in the latter. We evaluate the identification and verification performance of our proposed HAMFace on the large protocol of MegaFace.



(a) IJB-B



(b) IJB-C

Figure 9: ROC curves of IJB-B and IJB-C.

To conduct fair comparison with existing SOTA methods, HAMFace is trained on MS1MV3 with ResNet100 as backbone. Table 9 shows that our HAMFace gets the best result on these two general benchmarks. Figure 10 clearly demonstrates that our HAMFace is superior over the competitors on LFW and MegaFace benchmarks.

5. Conclusion

It is challenging to handle face recognition tasks with various intra-class variations in unconstrained scenarios in a unified manner. To address these problems, we have introduced a novel Hardness Adaptive Margin loss, which utilizes an adaptive sample-level margin according to the hardness levels of hard positive examples so that the face recognition performance on datasets with large intra-class variations can be improved. This gives our method a competitive advantage over other face recognition methods which target concrete variations. In essence, our proposed HAMFace adaptively assigns hard positive examples with larger margins and push them towards their corresponding class centers, which results in improved distribution of intra-class features and enhanced recognition performance. HAMFace

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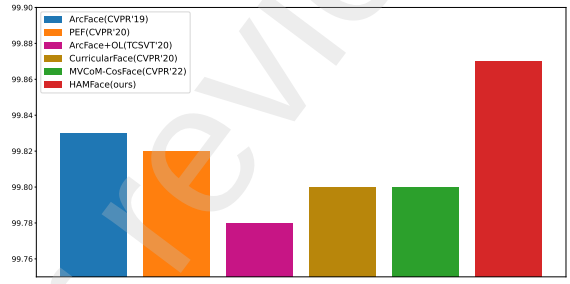
Table 8
VERIFICATION PERFORMANCE (%) ON IJB-B AND IJB-C.

Method	TAR on IJB-B			TAR on IJB-C		
	FAR=1e-6	FAR=1e-5	FAR=1e-4	FAR=1e-6	FAR=1e-5	FAR=1e-4
ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019)	38.28	89.33	94.25	86.25	93.15	95.65
CurricularFace (Y. Huang et al., 2020)	-	-	94.80	-	-	96.10
MagFace (Meng et al., 2021)	42.32	90.36	94.51	90.24	94.08	95.97
IHEM (Xiao et al., 2022)	39.79	91.37	95.04	90.93	**94.68**	96.46
HAMFace(our)	40.61	**91.56**	**95.04**	**91.25**	94.62	**96.46**

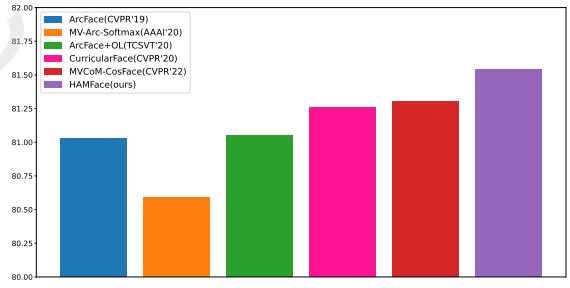
Table 9
COMPARISON RESULTS ON LFW AND MEGAFACE. 'MF1'REFERS TO THE MEGAFACE CHALLENGE 1 BENCHMARK. 'Id.' DENOTES THE RANK-1 IDENTIFICATION ACCURACY. 'Veri.' DENOTES THE TAR@FAR=1E-6.

Method	LFW	MF1
CosFace (H. Wang et al., 2018)	99.81	80.56
ArcFace (Deng, Guo, Xue, & Zafeiriou, 2019)	99.83	81.03
PFE (Shi & Jain, 2019)	99.82	78.95
MV-Arc-Softmax (X. Wang et al., 2020)	99.79	80.59
ArcFace+OL (Yang et al., 2020)	99.78	81.05
URL (Shi et al., 2020)	99.78	78.60
CurricularFace (Y. Huang et al., 2020)	99.80	81.26
MvCoM-CosFace (C. Liu et al., 2022)	99.80	81.30
HAMFace(ours)	**99.87**	**81.54**

achieved SOTA performance on several challenging benchmarks, even those with concrete variations and hybrid variations. Extensive experimental results have substantiated the superiority of our proposed HAMFace. In future studies, we will investigate how to extend HAMFace to other computer vision tasks, such as person re-identification and image retrieval, which also have the problem of intra-class variations.



(a) Verification performance (%) on LFW.



(b) RANK-1 Performance (%) on MegaFace Challenge 1.

Figure 10: Evaluation performance (%) on LFW and MegaFace Challenge 1 benchmarks.

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